# *Contents at a Glance*


***FOUNDATIONS***

Chapter 1. The Breadth and Depth of DSP . . . . . 1
Chapter 2. Statistics, Probability and Noise . . . . . 11
Chapter 3. ADC and DAC . . . . . 35
Chapter 4. DSP Software . . . . . 67

***FUNDAMENTALS***

Chapter 5. Linear Systems . . . . . 87
Chapter 6. Convolution . . . . . 107
Chapter 7. Properties of Convolution . . . . . 123
Chapter 8. The Discrete Fourier Transform . . . . . 141
Chapter 9. Applications of the DFT . . . . . 169
Chapter 10. Fourier Transform Properties . . . . . 185
Chapter 11. Fourier Transform Pairs . . . . . 209
Chapter 12. The Fast Fourier Transform . . . . . 225
Chapter 13. Continuous Signal Processing . . . . . 243

***DIGITAL FILTERS***

Chapter 14. Introduction to Digital Filters . . . . . 261
Chapter 15. Moving Average Filters . . . . . 277
Chapter 16. Windowed-Sinc Filters . . . . . 285
Chapter 17. Custom Filters . . . . . 297
Chapter 18. FFT Convolution . . . . . 311
Chapter 19. Recursive Filters . . . . . 319
Chapter 20. Chebyshev Filters . . . . . 333
Chapter 21. Filter Comparison . . . . . 343

***APPLICATIONS***

Chapter 22. Audio Processing . . . . . 351
Chapter 23. Image Formation and Display . . . . . 373
Chapter 24. Linear Image Processing . . . . . 397
Chapter 25. Special Imaging Techniques . . . . . 423
Chapter 26. Neural Networks (and more!) . . . . . 451
Chapter 27. Data Compression . . . . . 481
Chapter 28. Digital Signal Processors . . . . . 503
Chapter 29. Getting Started with DSPs . . . . . 535

***COMPLEX TECHNIQUES***

Chapter 30. Complex Numbers . . . . . 551
Chapter 31. The Complex Fourier Transform . . . . . 567
Chapter 32. The Laplace Transform . . . . . 581
Chapter 33. The z-Transform . . . . . 605

Glossary . . . . . 631
Index . . . . . 643